Inverse Mechanical Counting and Mathematics

The Axiom of 1, the Master Table, the Unified Operational Physics Engine, and the Inverse View from Closure

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The Axiom of 1

The Axiom. There is exactly one physical unit. This unit is the complete reality. All its equivalent views are

$$1 = -1 = \sqrt{1} = \sqrt{2^0}.$$

The square root belongs to the unit itself; it is not introduced from outside. The axiom is the starting point and the ending point of every mechanical reading. Every number that looks larger is a deeper inward reading of this same single unit; the axiom is the largest entity in the system, never exceeded.

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1 The Axiom of 1

There is one unit. Not one *of* something — one, and it is the whole. Everything that follows is this unit read inward, never anything built up toward it. The largest thing in the entire system is this 1; every number that looks larger is a deeper inward reading of it.

View	What it is	Coordinate
1	the unit, whole, at rest	$c = 0$, no mechanic
-1	the same unit read from across its own center	$c = 0$
$\sqrt{1}$	the unit carries its own root	$d = 0$
$\sqrt{2^0}$	the same statement in the master mechanic's writing	$d = 0$

Table 1: The four views of the one unit. All four are the same axiom of 1 at the same coordinate, $c = 0$, $d = 0$. The axiom is the floor and the whole of every reading.

2 The $\sqrt{2}$ Master Mechanic and the Master Table

The $\sqrt{2}$ mechanic is the master mechanic of the axiom — the simplest step beyond unity and the minimum step the axiom admits. The single unit is halved inward, count by count. One parity rule governs every reading: even operational depths d yield rational values; odd depths carry a natural factor of $\sqrt{2}$. Five active counts ($c = 1$ to 5) unfold nine active depths ($d = 1$ to 9). Every reading carries three coordinates — the mechanic $\sqrt{2}$, the depth d, and the count c. The full operational reading is the domains (2^c) times the unit ($\sqrt{2^d}$).

c	d	Mechanic $\sqrt{2^d}$	Unit decimal	Domains 2^c	Parity	Reading
0	0	$\sqrt{2^0} = 1$	1	1	even → rational	1
1	1	$\sqrt{2^1} = \sqrt{2}$	≈ 1.414	2	odd → irrational	$2\sqrt{2} \approx 2.828$
2	2	$\sqrt{2^2} = 2$	2	4	even → rational	8
2	3	$\sqrt{2^3} = 2\sqrt{2}$	≈ 2.828	4	odd → irrational	$8\sqrt{2} \approx 11.314$
3	4	$\sqrt{2^4} = 4$	4	8	even → rational	32
3	5	$\sqrt{2^5} = 4\sqrt{2}$	≈ 5.657	8	odd → irrational	$32\sqrt{2} \approx 45.255$
4	6	$\sqrt{2^6} = 8$	8	16	even → rational	128
4	7	$\sqrt{2^7} = 8\sqrt{2}$	≈ 11.314	16	odd → irrational	$128\sqrt{2} \approx 181.019$
5	8	$\sqrt{2^8} = 16$	16	32	even → rational	512
5	9	$\sqrt{2^9} = 16\sqrt{2}$	≈ 22.627	32	odd → irrational	$512\sqrt{2} \approx 724.077$

Table 2: The Master Table. The count c records the operation; the depth d records the master mechanic's exponent. Domains (2^c) times the unit ($\sqrt{2^d}$) gives the reading. The count closes at $c = 5$: domains 32, rational reading 512 at $d = 8$, irrational reading ≈ 724.077 at $d = 9$. The angular vector 720 is machined as $5 \times 9 \times 16$.

3 Where the 720 Is Machined From

The angular vector 720 is not assumed and not imported. It is machined directly from three quantities the master table already produces:

- the **5 active counts** — $c = 1, 2, 3, 4, 5$ (count 0 is the axiom at rest, no mechanic),

- the **9 active depths** — $d = 1, 2, 3, 4, 5, 6, 7, 8, 9$ (depth 0 is the axiom at rest), and
- the **terminal rational unit** 16 — $\sqrt{2^8}$, the even-depth ($d = 8$) unit at the closure count $c = 5$.

Multiplied together they give the angular whole:

$$720 = 5 \times 9 \times 16 = (\text{active counts}) \times (\text{active depths}) \times (\text{terminal rational unit}).$$

Quantity	Source on the master table	Value
Active counts	$c = 1, 2, 3, 4, 5$ (count 0 at rest)	5
Active depths	$d = 1, \ldots, 9$ (depth 0 at rest)	9
Terminal rational unit	$\sqrt{2^8}$, the $d = 8$ even unit at $c = 5$	16
Machined angular whole	$5 \times 9 \times 16$	720

Table 3: The 720 machined from the count and the depth.

The point of the machining is the bridge between the two tracks. At the closure count the rational face is $32 \times 16 = 512$ (domains times the $d = 8$ unit). The irrational face at the terminal depth $d = 9$ is $32 \times 16\sqrt{2} = 512\sqrt{2} \approx 724.077$, which runs into the endless decimals of $\sqrt{2^9} \approx 22.627$. Rather than carry that irrational tail, the machining pulls the solid rational coefficient 16 forward and multiplies it by the full structural span of the system — all 5 counts and all 9 depths — producing the crisp integer 720. This gives the angular face a stable, exact reference at the terminal boundary, and sphere over squared is the ratio of the two closures:

$$\frac{720}{512} = \frac{45}{32} = 1.40625.$$

4 The Base Units — Count 0, the Floor

Each base unit is one measurement, the axiom read one way:

Base unit	Measures	Track
gram (g)	mass / weight	rational (512)
meter (m)	length / distance	rational (512)
second (s)	time / cycle	angular (720)
coulomb (C)	charge	angular (720)
kelvin (K)	temperature	rational (512), via $T = \text{energy}/k$
mole (mol)	amount	rational (512)
candela (cd)	luminous intensity	fused: power (512) / solid angle (720)

Table 4: The seven base units at count 0. One base measurement each. Mass sits on the floor alongside length and time — not above it. Kelvin resolves to the rational track: temperature is energy/k, and energy is rational. Candela is genuinely fused — power (rational) over solid angle (angular).

5 How to Count

The count is the number of base measurements fused. Pieces double each count.

Count	Bases fused	Pieces	Name
0	1	1	floor (base unit)
1	2	2	dual
2	4	4	quadrant
3	8	8	octant
4	16	16	hexadecant
5	32	32	closure

Table 5: The counting rule. One base = count 0. Each fusion of further distinct base measurements raises the count; the pieces double.

6 The Operational Physics Quantities by Count

Quantity	Fuses	Standard	Count
length, mass, time, charge	base	m, g, s, C	0
velocity	length + time	m/s	1
frequency	cycle + time	1/s	1
area	length + length	m^2	1
current	charge + time	C/s = A	1
momentum	mass + velocity	g·m/s	2
acceleration	length + time + time	$\mathrm{m/s}^2$	2
voltage	current + resistance	V	2
energy	mass + velocity^2	$\mathrm{g{\cdot}m^2/s^2}$	3
force	mass + acceleration	$\mathrm{g{\cdot}m/s^2}$	3
pressure	force / area	$\mathrm{g/(m{\cdot}s^2)}$	3
power	energy + time	$\mathrm{g{\cdot}m^2/s^3}$	4
action ($\hbar$)	energy × time	$\mathrm{g{\cdot}m^2/s}$	4
G	volume / (mass·time^2)	$\mathrm{m^3/(g{\cdot}s^2)}$	5

Table 6: The operational unit counts for physics. The count is how many base measurements fuse. Power and action share count 4 — the time-mirror pair (energy per more time vs energy times time). G is the closure at count 5.

7 The Four Views Across Every Count (Forward)

Each count reads in four views on both tracks: rational $512/2^c$, angular $720/2^c$, the parity $\sqrt{2^c}$, and inverse parity $1/\sqrt{2^c}$.

8 The Inverse View from Closure — Three Branches

This is the count read backward from closure ($c = 5$) inward toward the axiom. Three branches run at every count — the rational ($512/2^c$), the spherical ($724.077/2^c$), and the angular reference ($720/2^c$). The mechanic preserves the large ratio $720/512 = 1.40625$ and the spherical lift 724.077/720 in inverse form.

c	Quantities at this count	Rational $512/2^c$	Angular $720/2^c$	$\sqrt{2^c}$	$1/\sqrt{2^c}$
0	base (g,m,s,C,K,mol,cd)	512	720	1.000000	1.000000
1	velocity, freq, area, current	256	360	1.414214	0.707107
2	momentum, accel, voltage	128	180	2.000000	0.500000
3	energy, force, pressure	64	90	2.828427	0.353553
4	power, action ($\hbar$)	32	45	4.000000	0.250000
5	G (closure)	16	22.5	5.656854	0.176777

Table 7: The four views across all counts. Forward (the seat), inverse (1/seat), parity ($\sqrt{2^c}$, the halving engine), inverse parity ($1/\sqrt{2^c}$). Rational quantities read the 512 column; temporal/cycle quantities read the 720 column; mixed quantities cross both.

- **Rational branch** — the 512 square closure, halved inward: $256, 128, 64, 32, 16$. The rational/displacement track.
- **Spherical branch** — the 724.077 irrational sphere closure ($512\sqrt{2}$), halved inward: $362.04, 181.02, 90.51, 45.25, 22.63$. The $\sqrt{2}$ lift track.
- **Angular reference** — the 720 machined cycle, halved inward: $360, 180, 90, 45, 22.5$. The partition line every reading reconciles to.

c	Rational $512/2^c$	1/Rat	Spherical $724.077/2^c$	1/Sph	Angular $720/2^c$	1/Ang
0	512	0.001953	724.0773	0.001381	720	0.001389
1	256	0.003906	362.0387	0.002762	360	0.002778
2	128	0.007813	181.0193	0.005524	180	0.005556
3	64	0.015625	90.5097	0.011049	90	0.011111
4	32	0.031250	45.2548	0.022097	45	0.022222
5	16	0.062500	22.6274	0.044194	22.5	0.044444

Table 8: The inverse master table, read from the closures inward. Three branches at every count, each in forward and inverse view. All values exact: the rational is $512/2^c$, the spherical is $512\sqrt{2}/2^c$, the angular is $720/2^c$.

c	Rational inv	Spherical inv	Angular inv	$\sqrt{2^c}$	$1/\sqrt{2^c}$
0	0.001953	0.001381	0.001389	1.000000	1.000000
1	0.003906	0.002762	0.002778	1.414214	0.707107
2	0.007813	0.005524	0.005556	2.000000	0.500000
3	0.015625	0.011049	0.011111	2.828427	0.353553
4	0.031250	0.022097	0.022222	4.000000	0.250000
5	0.062500	0.044194	0.044444	5.656854	0.176777

Table 9: The inverse four-view readings. The inverse branches are the natural inward magnitudes; the parity $\sqrt{2^c}$ and its inverse are the halving engine read at each count. Rational, spherical, and angular each give a distinct, exact measurement.

9 Physics Quantities in the Inverse View

The physics quantities sit at their fusion count and read inward on the branch their character selects. Mass lives completely on the floor (count 0) as a foundational base measurement.

c	Quantities	Rational inv	Spherical inv	Angular inv
0	base (g, m, s, C)	0.001953	0.001381	0.001389
1	velocity, freq, area, current	0.003906	0.002762	0.002778
2	momentum, accel, voltage	0.007813	0.005524	0.005556
3	energy, force, pressure	0.015625	0.011049	0.011111
4	power, action ($\hbar$)	0.031250	0.022097	0.022222
5	G (closure)	0.062500	0.044194	0.044444

Table 10: Physics quantities in the inverse view, each count reading inward on all three branches. The fusion count fixes the row; the branch fixes which inverse measurement the quantity reads.

10 The Constant Coordinates

Each constant sits at its count as a coordinate of the lattice, read on its branch.

Constant	Count / path	Rational inv	Spherical inv	Angular inv
Speed of light c	count 1 dual	0.003906	0.002762	0.002778
Fine-structure 1/137	count 2: $\pm(128 + 9)$	0.007813	0.005524	0.005556
Fine-structure 1/137	count 4: $\pm(4 \cdot 32 + 9)$	0.031250	0.022097	0.022222
Machine $\pi = 201/64$	count 3 octant	0.015625	0.011049	0.011111
Action $\hbar$	count 4 sector	0.031250	0.022097	0.022222
G	count 5 closure	0.062500	0.044194	0.044444
Boltzmann k	count 3 (energy octant)	0.015625	0.011049	0.011111

Table 11: The constant coordinates on the lattice. Each constant is seated at its fusion count and read inward on the three branches. The values are exact coordinates of the lattice (fractions of the closures), produced by the count and the master mechanic alone.

11 Two Kinds of π — The Anchor and the Process

Machine $\pi = 201/64 = 3.140625$ is a fixed structural anchor at count 3 — one rational value, no decimals, a locked gear that never moves. It does not approximate anything; it is the count-3 coordinate $(3 \times 64 + 9)/64$, and it holds the lattice in place.

The polygon track is the open process — it runs as far as the count goes ($c = 5$, 6, 12, 100, and beyond), each step a deeper measurement revealing more digits of standard π. It never has to stop.

The anchor holds the lattice; the track measures the continuum. They are two different objects, not two estimates of the same thing. The two sections that follow give the track in detail.

12 The Extended Polygon Track — Machine π and the Continuum

Machine $\pi = 201/64 = 3.140625$ is the fixed rational anchor at count 3. The continuous circle can also be read as an inscribed-polygon sequence: an n-gon with $n = 2^c$ sides estimates π. This sequence climbs upward toward standard π as the count rises, and it crosses the machine-π anchor between counts 6 and 7.

Count	Sides 2^c	Angular partition $720/2^c$	π estimate
5	32	22.5°	3.136548
6	64	11.25°	3.140331
7	128	5.625°	3.141277
8	256	2.8125°	3.141514
9	512	1.40625°	3.141573
10	1024	0.703125°	3.141588
11	2048	0.3515625°	3.141591
12	4096	0.17578125°	3.141592

Table 12: The extended polygon track. Each π estimate is the value of the 2^c-gon (verified exact). The sequence climbs toward standard π; machine $\pi = 201/64 = 3.140625$ sits between count 6 (3.140331) and count 7 (3.141277). At count 9 the 512-gon's side count matches the base rational seat 512, and the angular partition 1.40625° equals the machining ratio 720/512.

This is the difference between the two readings of π. The polygon track is the open process: to reach a perfectly smooth circle it must count outward toward infinity ($c \to \infty$), gaining a finer decimal at every step. Machine π (201/64) drops a fixed rational anchor at count 3, the closed structural value that holds the lattice together. The polygon track shows where that anchor sits relative to the continuum — crossed early, between counts 6 and 7 — while the lattice itself closes at count 5.

13 The Repeating Count — One Mechanic, Every Octave

The count closes at 5, but the mechanic does not end. The axiom of 1 and the $\sqrt{2}$ master mechanic simply repeat on the next octave: the same five-count cycle fires again at a deeper scale, count 6 through count 10 being the second octave, count 11 through 15 the third, and so on. The engine never changes — it is the one axiom, halved by the one mechanic, looping down the lattice.

This is why the polygon track climbs toward standard π without ever needing a new rule. Each count is one more repetition of the $c = 1$ halving on a finer layer ($n = 2^c$ sides), and standard π is reached in that repetition by **count** 12 (the 4096-gon, 3.141592). What standard mathematics calls the "endless decimals" of π is not chaos — it is the first count repeating its motion forever, one mechanic looping down the scale. Machine $\pi = 201/64$ drops the fixed anchor at count 3 of the first octave; the continuum value is simply where the same repeating count arrives after twelve firings.

14 The Full 12-Count π Sequence

The repeating count is driven by the polygon: an n-gon with $n = 2^c$ sides, the same $c = 1$ halving applied c times. Counting straight through the closure wall at $c = 5$ and on to $c = 12$,

Octave	Counts	Role
First	$c = 1$ to 5	the axiom unfolds to closure (512, 720, machine π at $c = 3$)
Second	$c = 6$ to 10	the same mechanic repeats; polygon track climbs past machine π
Third	$c = 11$ onward	the repetition continues; standard π reached at $c = 12$

Table 13: The repeating count. The five-count cycle of the axiom and the master mechanic repeats octave by octave. Standard π (3.141592) is reached on the polygon track at count 12, the same first count repeating its halving down the scale — not a new rule, the one mechanic looping forever.

the value climbs toward standard π as the machine fires the identical operation on ever-finer scales.

c	Sides 2^c	π_c
1	2	2.0000000000
2	4	2.8284271247
3	8	3.0614674589
4	16	3.1214451523
5	32	3.1365484905
6	64	3.1403311570
7	128	3.1412772509
8	256	3.1415138011
9	512	3.1415729404
10	1024	3.1415877253
11	2048	3.1415914215
12	4096	3.1415923456

Table 14: The full 12-count π sequence, every value of the 2^c-gon (verified exact to ten places). The first five counts are the primary cycle to closure; counts 6 through 12 repeat the same halving on the next octaves. Machine $\pi = 201/64 = 3.140625$ is crossed between count 6 (3.140331) and count 7 (3.141277). Standard π (3.141592) is reached at count 12.

The value of π is read as a function of the count. At low counts ($c = 1$ to 5) the measurement is rough — few large domains. Through closure ($c = 6$ to 12) the machine does not stop; it repeats the same operational halving on smaller scales, each added count a deeper measurement revealing the next digits. The infinite continuum maps to a finite counting scale: the same first count, repeating its motion, arriving at standard π by count 12.

15 The 4D Inverse Operational Tensor — Count 1 to Closure

The same five counts, read as the inverse operational tensor: each count switches on one more dimensional index, building from a 1D dual link to full 4D structural closure. The squared face steps $256 \to 128 \to 64 \to 32 \to 16$ ($512/2^c$); the sphere face steps $360° \to 180° \to 90° \to 45° \to 22.5°$ ($720/2^c$); the parity engine steps $\sqrt{2^c}$. The coupling $720/512 = 1.40625$ stays anchored throughout.

At closure ($c = 5$) the saturated 4D tensor components are the count-5 inverse units raised to the fourth (dimensional) power: the squared hyper-amplitude $1/128^4 = 1/268{,}435{,}456$ and

c	Count stage	Squared seat	Sphere seat	Parity $\sqrt{2^c}$	Tensor state
1	Dual split	256 ($-256/+256$)	360° (±360°)	1.414214	1D linear vector links
2	quadrant (count 2)	128	180°	2.000000	2D matrix projection (Schrödinger)
3	octant (count 3)	64	90°	2.828427	3D volumetric grading (machine π)
4	hexadecant (count 4)	32	45°	4.000000	4D relativistic tensors ($\hbar$)
5	closure (count 5)	16	22.5°	5.656854	full closure (G)

Table 15: The 4D inverse operational tensor progression. Each count activates one more count stage (count 2 through count 5), building dimensional density from a 1D dual link to the sealed 4D closure. Squared seats are $512/2^c$, sphere seats $720/2^c$ — the correct per-count stepping, consistent with the master table.

the sphere hyper-phase $1/180^4 = 1/1{,}049{,}760{,}000$. The tensor locks inward at the closure count, forming the sealed terminal boundary of the count — the gravitational closure at count 5, where the count has no further operational room to unfold.

16 Boltzmann's Constant and Temperature on the Count

This section seats the last constant, Boltzmann's k, and traces temperature across all five counts to the terminal CMB and closure boundaries.

16.1 Boltzmann's Constant k

In standard physics $k = 1.380649 \times 10^{-23}$ J/K (exact since the 2019 SI revision). Its units J/K = $(\text{g}\cdot\text{m}^2/\text{s}^2)$/K make it the energy–temperature conversion: it carries the count-3 energy octant as a per-temperature ratio, so k seats at **count** 3. Temperature is then energy with k divided out, $T = \text{energy}/k$, and the $\frac{3}{2}$ in KE $= \frac{3}{2}kT$ is the three octant directions over the count-1 dual.

16.2 The Thermodynamic Mapping Across the Counts

Count 0 — Ground state kelvin. The static floor: absolute zero (0 K). No kinetic displacement, no phase rotation. Kelvin rests as an un-excited base dimension.

Count 1 — Dual thermal gradient. First thermal halving; seats 256 / 360. A 1D linear heat gradient flowing between the dual boundaries -256 and $+256$. Thermal displacement and phase expansion are separate channels.

Count 2 — Quadrant thermal plane. Seats 128 / 180. Heat expands into a 2D plane, coupled to orthogonal phase rotation; planar conduction, $k\,T_2$ reading as the momentum/voltage quadrant.

Count 3 — Octant thermodynamic field (the home of k). Seats 64 / 90. Boltzmann's k seats here. The four views for k: rational inverse 0.015625, spherical inverse 0.011049, angular inverse 0.011111. The operational reading is

$$\text{KE} = \left(\frac{\text{3 octant directions}}{\text{count-1 dual}}\right) kT_3 = \tfrac{3}{2}\,kT_3, \qquad T_3 = \frac{\text{energy}}{k}.$$

Heat is 3D fluidic volume; temperature measures the volumetric kinetic energy density across the 8 octant pieces.

Count 4 — Hexadecant radiation field (the CMB anchor). Seats 32 / 45. The domain of action ($\hbar$) and the relativistic photon gas. The CMB reads here:

$$T_{\text{CMB}} = \frac{264\sqrt{2}}{137} \approx 2.72519 \text{ K},$$

against measured 2.72548 K — 0.011%. The background temperature is read as a fixed structural 4D radiation coordinate dictated by the 137 fine-structure corridor on the count-4 tier.

Count 5 — Thermodynamic closure. Terminal seats 16 / 22.5°; saturated units $1/128^4$ and $1/180^4$. Heat runs out of degrees of freedom; temperature and entropy compress into the count-5 closure boundary and lock inward, no leakage out of the closed 32-domain phase space.

16.3 The Thermodynamic Progression Matrix

Count	Rational seat	Angular seat	Inverse parity	Thermodynamic state
$c = 0$	512	720	1.000000	Absolute zero ground state (the floor)
$c = 1$	256	360	0.707107	1D linear heat gradient (velocity track)
$c = 2$	128	180	0.500000	2D orthogonal conduction matrix
$c = 3$	64	90	0.353553	3D volumetric energy (seat of k and $\frac{3}{2}$)
$c = 4$	32	45	0.250000	4D relativistic radiation (CMB = 2.72519 K)
$c = 5$	16	22.5	0.176777	Closed gravitational thermal closure (G)

Table 16: The thermodynamic progression. The machining ratio $720/512 = 1.40625$ stays active across all tiers; thermal conversion is synchronized with spatial displacement and phase rotation. k seats at count 3; the CMB reads at count 4 through the 137 corridor; closure at count 5.

17 Quantum Gravity as Natural Closure at Count 5

At Count 5 the system reaches full saturation. All four count stages (counts 2, 3, 4, 5) have fired. The 4D inverse operational tensor is now complete on both faces and locks into the sealed closure boundary. This closure boundary **is** quantum gravity in this framework.

At Count 5 the inverse units reach their terminal power:

- Squared face (hyper-amplitude): $1/128^4$
- Sphere face (hyper-phase): $1/180^{4\circ}$

The wavefunction at closure (ψ_5) no longer evolves. Kinetic displacement on the squared face and phase rotation on the sphere face reach perfect mutual cancellation. The system becomes a static containment structure. The mechanical statement at closure is:

$$\left(\frac{720}{512}\right)^2 \frac{1}{2m}\,\partial^2\psi_5 + G_{\text{closure}} \cdot |\psi_5|^2\psi_5 = 0$$

Where:

- The ratio $720/512 = 1.40625$ remains the fixed coupling between faces.
- The second derivative term has been fully countered by the terminal inverse unit scale.
- G_{closure} is the self-containment strength read directly from the Count 5 angular seat (22.5°) and the rational seat 16.

This is not an added force. It is the geometric pressure that appears when the count has no further operational room to unfold. The rational 512 boundary and the spherical 724.077 boundary have compressed into each other completely. The result is the sealed terminal closure of the count — the point at which the count $0, 1, 2, 3, 4, 5$ reading inward has no further operational room to unfold.

The irrational sphere closure at terminal depth $d = 9$ confirms the geometry:

$$512\sqrt{2} \approx 724.077 \implies \frac{724.077}{512} = \sqrt{2}$$

Gravity, in this reading, is the final geometric reconciliation of all active domains back to the single Axiom of 1.

Realization. The whole engine runs on one set of gears. From the axiom of 1 and the $\sqrt{2}$ master mechanic, the master table unfolds five counts and nine depths to closure at 512 (rational) and 724.077 (spherical), with 720 the machined angular reference. The base units sit on the floor at count 0; the count is how many base measurements fuse; the four views (forward, inverse, parity, inverse parity) read each count on the rational, spherical, and angular branches. Read outward it is the forward physics engine; read inward from closure it is the inverse view. The mathematics is the reading; the single physical unit is the thing read.